

Manual

Discrete Consumption Recording AIP-DVE 8.2

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1 Overview of Discrete Consumption Recording

Purpose

Key input values when collecting shop floor data are times and quantities. While times or durations are used to describe the time effort that was required to manufacture a material, quantities document the entire scope of the produced material. The objective is to process or output the material quantities defined in the order or the operation.

In most cases, the quantities entered are sufficient to be able to execute the actions that will change stock quantities in the upper-level ERP system accordingly.

- Goods receipt from production
The finished quantity will increase stock in the ERP system.
- Goods issue from production
Based on the finished quantity, the consumption of the material that flowed in (so-called input material) can be calculated in reverse order of the production process in the ERP system, accounting for the bill of materials the order is based on. This will lead to a reduction of stock in the ERP system.

However, this kind of consumption calculation is oftentimes not enough to ensure a "clean" inventory management in the upper-level ERP system. Instead, there is a need to discretely enter order-related material consumption and to then post the consumption later in the ERP system.

Implementation considerations

You make use of this functions package if you would like to enter material consumption discretely and without a batch reference and upload it to an inventory management system.

Integration

The function can be integrated into a system dedicated to shop floor data collection (BDE), or also be used within the context of material and production logistics (MPL/ TRT).

Features

- Functions for discretely entering order-related material consumptions without a batch reference.
- Posting dialog at the Windows Terminal AIP to input discrete consumptions relating to material components during operation logoff, OP interruption or partial confirmation/upload.
- Entry based on the produced and manually entered total quantity or yield and/ or scrap.

- Providing material consumptions for confirmation/upload from HYDRA to the inventory management system in HYDRA standard format (requires that the interface used to upload material and batch data is licensed and activated).



Posting dialogs are activated during HYDRA customizing (a chargeable service).

2 Discrete Consumption Input in AIP

Usage

The function described below makes it possible to enter material consumption at the AIP shop floor terminal so that it can be uploaded via an interface in the form of goods movements.

Integration

Material consumption input can take place at the AIP shop floor terminal while an operation is being interrupted, while an operation is being logged off or when entering a partial confirmation/upload.

Requirement

The relevant utilization material must be assigned to the operation in order to discretely record consumption. The meaning of the following fields must be observed in particular (logical consideration). The material components can be assigned in HYDRA manually ([Order management](#) → [Edit components](#)) or from the upper level system.

Material number

The material number uniquely identifies a material. In the component list, a material is only unique in association with a BOM item.

BOM item

The BOM item combined with the material number forms a unique key for a material component at an operation.

Material designation

The material designation is used to describe a material more exactly. It functions as commentary and is displayed in the input dialog.

Material category

The material category controls, among other things, the way a material is technically processed in the HYDRA system. Relating to the material consumption input (not batch-related), the following value is important:

"M" Material component, should be entered in terms of consumption

Material type

The material type is another parameter used for processing control in HYDRA. The material type is used to control whether goods movement created from the material consumption should be uploaded to the ERP system or not. Unless defined otherwise assign the material type SYSTEM here.

To assure that a goods issue is uploaded via the interface, the option *Transfer to interface* must be set at the material type that the utilization material is based on.

When launching HYDRA for the first time or during discrete consumption input, it will need to be coordinated whether or for which material components consumption should be uploaded to the ERP system. If the MPL module is not used, this configuration is made during the HYDRA customizing process.

Consumption type

Set the consumption type to "D" (discrete consumption input) for material components that this function is used for to collect material consumption.



It is possible to transfer the consumption type using the relevant component segment of the ERP interface (EIS-ERP). When using the HYINFO interface as a part of the PP-PDC interface, the consumption type is made available by customizing the system accordingly.

The interface necessary to upload material consumption is not a part of this function.

Input quantity

The input quantity is a component quantity needed to manufacture one unit (one piece, for example) of the output material (article) being produced. It is used to calculate the theoretical material consumption based on the produced quantity.

Unit

Quantity unit of the material component in which material consumption is recorded.

Entering material consumption at AIP



The customer system must be customized accordingly before the functions described below become available.

The following functions are available at the AIP for entering material consumption based on the produced (total) quantity.

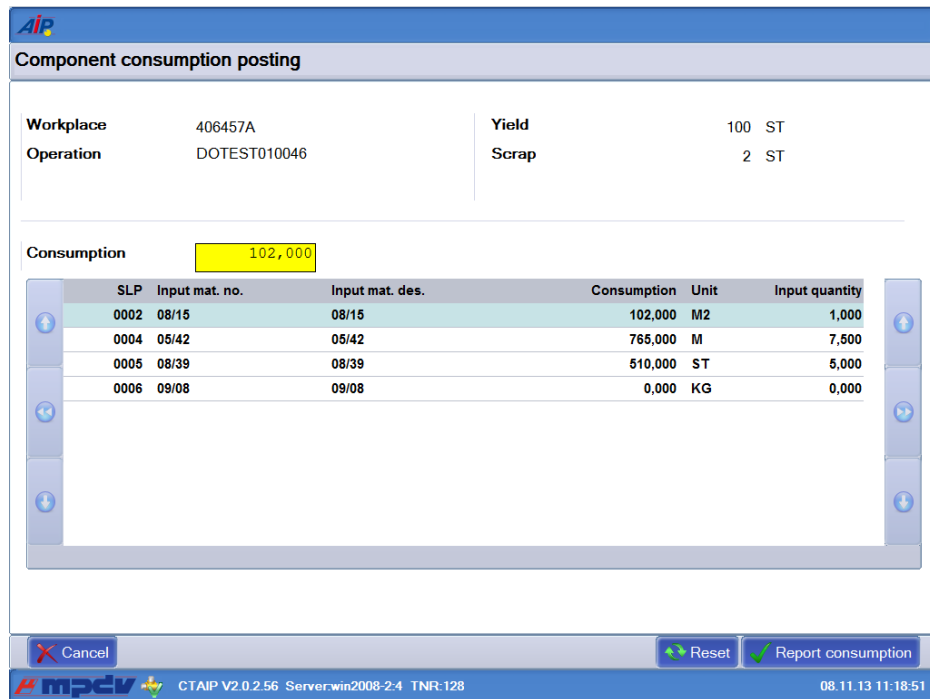
In the dialogs

- Log operation off (A_AB)
- Interrupt operation (A_UN)
- Partial confirmation/upload for operation (A_TR)

an additional *consumption* button is available with which the *Component consumption posting* dialog can be called up.

It is also possible to make the button available that is used to call up the *Component consumption posting* dialog from the MPL specific dialogs used for operation interruption (A_UN_MPL, A_UN_RF, A_UN_RS) or for operation log off (A_AB_MPL, A_AB_RF, A_AB_RS). However, what needs to be considered in this regard is that in MPL, consumption of batch-related material is posted differently.

The input dialog opens after the *Consumption* button is pressed:



SLP	Input mat. no.	Input mat. des.	Consumption	Unit	Input quantity
0002	08/15	08/15	102,000	M2	1,000
0004	05/42	05/42	765,000	M	7,500
0005	08/39	08/39	510,000	ST	5,000
0006	09/08	09/08	0,000	KG	0,000

"Component consumption posting (A_VERB)" dialog in AIP

The status information shown includes the workplace displayed in the dialog called up, the operation and also the yield and scrap quantities that are also entered in the dialog that is called up. When calling up from an MPL dialog, the entered quantity is displayed based on whether classified ("Quality") as scrap or as yield.

Furthermore, a table is displayed showing the component that is flagged in the component list as consumption type "D". The following data are displayed in the table:

BOM item

BOM item

Input.Mat.No.

Material number of the material component.

Input.Mat.Des

Material designation of the material component.

Consumption

Calculated consumption based on the quantity entered in the dialog called up:

$$\text{Calculated consumption} = (\text{yield} + \text{scrap}) * \text{input quantity}$$



Only yield and scrap quantities are taken into accounts when calculating the consumption.

Quantities are not set off against each other (e.g. scrap set off against yield) when the calculated consumption is determined.

Unit

Quantity unit (unit of the input quantity)

Input quantity

Input quantity required to produce one quantity unit of the output material.

When a component is selected from the list, the calculated consumption is proposed in the *Consumption* input field. If the actual consumption deviates from the calculated consumption, the operator can now modify this quantity; it is transferred to the list. Any consumption that was modified manually is shown highlighted in "green" in the list (e.g. ).

By pressing the *Reset* key, the component list can be called up again; this will recalculate the consumption quantities; any consumption quantities that were already modified will be overwritten.

After confirming this dialog and the dialog that is called up, the consumption quantities are updated and are thus transferred to the server via the posting. There, the consumption is updated as the status in the component list at the operation. In addition, one material movement (goods issue) is written for each component.



The material consumption is only posted as a goods movement if the dialog was called up explicitly. Material consumptions that were not posted are not automatically posted when the operation is logged off or interrupted.

If the consumption input dialog is integrated in the partial confirmation dialog and if it is not called up when a partial confirmation is executed or when the entry is interrupted, the system will remember the calculated consumption for all material components of the consumption type "D" and will account for it the next time the consumption input dialog is called up.

Example (assumption: a material component with the input quantity 2 is defined at the operation):

- Call up the partial confirmation (A_TR): Enter yield 9, scrap 1
- Call up the consumption input dialog (A_VERB)
 - $(9 * 2 + 1 * 2 =) 20$ is proposed as material consumption

- Cancel the dialog (do not click on OK to confirm)
- Confirm the partial confirmation/upload dialog (A_TR)
 - The yield 9 and the scrap 1 are posted at the operation.
 - The system remembers the calculated consumption of 20
- Call up the partial confirmation (A_TR): Enter yield of 5
- Call up the consumption input dialog (A_VERB)
 - $(20 + 5 * 2 =)$ 30 is now proposed as material consumption
 - Confirm with OK
- Confirm the partial confirmation/upload dialog (A_TR)
 - The yield 5 is posted to the operation.
 - The consumption of 30 is posted as a goods movement.

If after calling up the consumption input dialog it is canceled and thus closed and the quantity is then modified in the dialog that appears and if then the consumption input dialog is once again called up, then the consumption quantities that were proposed the first time it was called up will still be shown in this dialog. This is why the proposed material consumption will need to be updated by clicking on the "Reset" button as the case may be.

Please note with regard to customizing

In addition to making the necessary programs available, please also consider the following

- Activate the dynamic dialog A-VERB, accounting for existing terminal groups in some cases.
- Integrate the button used to call up the consumption input dialog in the relevant posting dialogs. The following configurations are possible here, whereas a. can be seen as an alternative to b. and c.:
 - a. Integration into the workflow step WF_AA_QUA.
The button is found in the standard dialogs
 - Interrupt OP (A_UN),
 - Log off OP (A-AB), and
 - Partial confirmation/upload (A_TR)
 - in each workflow step in which quantities are entered manually.
 - b. Integration into the workflow dialog WF_AUN_CHK
The button is available in the standard dialog Interrupt OP (A_UN) in the confirmation workflow step.
 - c. Integration into the workflow dialog WF_AAB_CHK
The button is available in the standard dialog Log off OP (A-AB) in the confirmation workflow step.
- Integrating or checking the grid configuration for the dialog A_VERB (ctaipay.ini)
- Optional: Uploading the material consumption as goods movement

- Determining which material type is transferred to the material component from the ERP system (typically SYSTEM)
- Setting the flag "Transfer to interface" at the material type. Options:
 - Directly via the master data configuration (only possible, if MPL/ TRT is active)
 - Per SQL to the HYDRA server (mysql -r -):
update hz_typen set we_ext_kz = 'J' where hz_typ = 'SYSTEM';
 - Per SQL via the SQL tester in MOC:
update hz_typen set we_ext_kz = 'J' where hz_typ = 'SYSTEM'
 - Per BAPI to the HYDRA server (Observe system!):
hymwb -u9999 -
c"DLG=MATTYP.UPDATE|MATTYP.MATTYP=SYSTEM|MATTYP.WEEXT:GEN
=J|DAT=today|ZEI=now|"
- Check per SQL: select hz_typ, we_ext_kz from hz_typen
- Activate the material upload interface (see relevant documentation).